

Brutal Practice Paper B21 - Forests, Large Numbers, Weather & Climate, A Peek Beyond the Point

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. Sunlight falls on a forest. The canopy lets through only 20% of it, the shrub layer below lets through 25% of whatever reaches it, and the herb layer lets through 40% of whatever reaches it. What percent of the original sunlight finally reaches the forest floor? [\[Forests\]](#)

- (A) 5%
- (B) 1%
- (C) 20%
- (D) 10%
- (E) 8%
- (F) 2.5%
- (G) 2%

2. In the number 6,47,80,521, look at the digit 7. What is the difference between its place value and its face value? [\[Large Numbers\]](#)

- (A) 6,99,000
- (B) 69,993
- (C) 6,99,933
- (D) 9,93
- (E) 70,000
- (F) 6,99,993
- (G) 7,00,000

3. On one day a town's highest temperature is 34.5 deg C and its lowest is 22.8 deg C. What is the day's temperature range (highest minus lowest)? [\[Weather, Climate & Adaptations\]](#)

- (A) 11.7 deg C
- (B) 12.3 deg C
- (C) 57.3 deg C
- (D) 13.7 deg C
- (E) 11.6 deg C
- (F) 11.3 deg C

4. Look at the decimal number 63.475. What is the place value of the digit 7 in it? [\[A Peek Beyond the Point\]](#)

- (A) 0.7
- (B) 70
- (C) 7.0
- (D) 0.07
- (E) 7
- (F) 0.007
- (G) 0.7 hundredths

5. In one year a patch of forest drops 6000 kg of dead leaves. Decomposers turn $\frac{5}{8}$ of this into humus and the rest escapes as gases. How many kilograms become humus, and how many kilograms escape as gases? [\[Forests\]](#)

- (A) 3600 kg humus, 2400 kg gases
- (B) 3000 kg humus, 3000 kg gases
- (C) 3750 kg humus, 2400 kg gases
- (D) 3750 kg humus, 2250 kg gases
- (E) 2250 kg humus, 3750 kg gases
- (F) 4800 kg humus, 1200 kg gases

6. Round 4,86,729 to the nearest thousand, and also round it to the nearest ten-thousand. What is the difference between these two rounded values? [\[Large Numbers\]](#)

- (A) 13,000
- (B) 4,000
- (C) 10,000
- (D) 3,000
- (E) 7,000
- (F) 30,000

7. Over five days a town records rainfall of 12 mm, 0 mm, 8 mm, 15 mm and 5 mm. What is the mean (average) daily rainfall over these five days? [\[Weather, Climate & Adaptations\]](#)

- (A) 8.5 mm
- (B) 8 mm
- (C) 9 mm
- (D) 10 mm
- (E) 6 mm
- (F) 40 mm
- (G) 7.5 mm

8. What is the value of 2.5×0.04 ? [\[A Peek Beyond the Point\]](#)

- (A) 0.4
- (B) 0.1
- (C) 1.0
- (D) 0.001
- (E) 0.01
- (F) 0.25

9. In a forest food chain, grass holds 10000 energy units. A deer that eats grass keeps only 10% of the energy it takes in, and a tiger that eats the deer again keeps only 10%. How many energy units finally reach the tiger? [\[Forests\]](#)

- (A) 50 units
- (B) 100 units
- (C) 2000 units
- (D) 1000 units
- (E) 200 units
- (F) 10 units
- (G) 500 units

- 10.** A toy factory makes 2475 toys every day. How many toys does it make in one ordinary (non-leap) year of 365 days? [\[Large Numbers\]](#)
- (A) 8,03,375
 - (B) 9,03,275
 - (C) 9,04,375
 - (D) 9,02,375
 - (E) 9,03,375
 - (F) 9,90,000
 - (G) 8,93,375
- 11.** Which one of these statements describes the CLIMATE of a place rather than its weather on a single day? [\[Weather, Climate & Adaptations\]](#)
- (A) The sky is clear and sunny at this moment
 - (B) Thick fog covered the valley this morning
 - (C) Over many years this place is generally hot and receives heavy rain from June to September
 - (D) It is raining heavily here right now
 - (E) A thunderstorm is expected later tonight
 - (F) A cold wind is blowing through the town today
 - (G) The temperature dropped sharply this afternoon
- 12.** What is 0.144 divided by 1.2? [\[A Peek Beyond the Point\]](#)
- (A) 0.012
 - (B) 1.12
 - (C) 0.0012
 - (D) 1.2
 - (E) 0.12
 - (F) 12
 - (G) 0.6
- 13.** A grown tree releases about 120 kg of oxygen in a year. A small community forest has 250 such trees. If one person needs about 600 kg of oxygen in a year, how many people can this forest's yearly oxygen support? [\[Forests\]](#)
- (A) 30 people
 - (B) 40 people
 - (C) 100 people
 - (D) 50 people
 - (E) 25 people
 - (F) 60 people
- 14.** What do you get when you subtract the smallest 6-digit number from the greatest 7-digit number? [\[Large Numbers\]](#)
- (A) 9,99,999
 - (B) 99,00,000
 - (C) 89,99,999
 - (D) 99,99,899
 - (E) 98,99,999
 - (F) 98,99,899
 - (G) 9,89,999

15. On a humid day sweat does not dry off our skin easily and we feel uncomfortable. What does high humidity mean, and why does it make us feel hotter? [\[Weather, Climate & Adaptations\]](#)

- (A) Humidity is the amount of dust floating in the air
- (B) High humidity means there is no water vapour in the air
- (C) Humidity means the air is colder than usual
- (D) Humidity makes sweat evaporate faster, cooling us more
- (E) The air already holds a lot of water vapour, so our sweat evaporates slowly and cools us less
- (F) High humidity means strong winds are blowing

16. Rice costs Rs 48.50 per kilogram. What is the cost of 3.5 kilograms of this rice? [\[A Peek Beyond the Point\]](#)

- (A) Rs 170.75
- (B) Rs 172.50
- (C) Rs 16.975
- (D) Rs 169.50
- (E) Rs 145.50
- (F) Rs 169.75
- (G) Rs 168.75

17. Which arrangement lists the layers of a forest correctly from the tallest at the top to the shortest at the bottom? [\[Forests\]](#)

- (A) Canopy, shrub layer, understorey, herb layer
- (B) Herb layer, canopy, shrub layer, understorey
- (C) Canopy, understorey, shrub layer, herb layer
- (D) Understorey, canopy, herb layer, shrub layer
- (E) Canopy, herb layer, shrub layer, understorey
- (F) Herb layer, shrub layer, understorey, canopy
- (G) Shrub layer, herb layer, canopy, understorey

18. Using each of the digits 3, 0, 5, 8 and 1 exactly once, form the greatest possible number and the smallest possible number (no number may begin with 0). What is their difference? [\[Large Numbers\]](#)

- (A) 74,942
- (B) 74,852
- (C) 75,042
- (D) 84,952
- (E) 73,952
- (F) 74,952
- (G) 75,952

19. A polar bear has thick white fur and a deep layer of fat under its skin. Which of these is NOT a reason these features help it survive in the freezing Arctic? [\[Weather, Climate & Adaptations\]](#)

- (A) The fat helps it lose body heat quickly to the cold air
- (B) The fat layer insulates the body against the cold
- (C) The thick fur keeps body heat from escaping
- (D) The fur traps a layer of warm air near the skin
- (E) The white fur camouflages it against the snow
- (F) The fat is also a store of food energy
- (G) The white colour makes it hard for prey to spot

20. Which of these decimals is the largest: 0.5, 0.45, 0.405, 0.045, 0.54? [\[A Peek Beyond the Point\]](#)

- (A) 0.405 is largest
- (B) 0.045
- (C) 0.54
- (D) 0.5
- (E) 0.45
- (F) they are all equal
- (G) 0.405

21. A 2 kg sample of forest soil is 15% humus by weight. This humus can soak up water equal to 3 times its own weight. What is the greatest mass of water this sample's humus can hold? [\[Forests\]](#)

- (A) 0.9 kg
- (B) 0.3 kg
- (C) 1.2 kg
- (D) 6 kg
- (E) 0.45 kg
- (F) 0.6 kg

22. A water tank holds 12,50,000 litres when full and starts full. A town draws 8,75,400 litres from it. Then 2,13,750 litres of fresh water are added. How many litres are now in the tank? [\[Large Numbers\]](#)

- (A) 1,61,650
- (B) 5,88,350
- (C) 13,38,350
- (D) 6,88,350
- (E) 5,87,350
- (F) 5,88,300
- (G) 5,88,450

23. Emperor penguins huddle together in groups to stay warm. A colony has 1200 penguins, and each tight huddle is made of exactly 40 penguins. How many huddles does the colony form? [\[Weather, Climate & Adaptations\]](#)

- (A) 32
- (B) 36
- (C) 24
- (D) 300
- (E) 48
- (F) 30

24. Add the three decimals $12.07 + 9.9 + 0.123$. What is the sum? [\[A Peek Beyond the Point\]](#)

- (A) 22.93
- (B) 22.093
- (C) 22.193
- (D) 13.183
- (E) 21.093
- (F) 23.093

25. A growing tree takes in about 22 kg of carbon dioxide in a year. A car gives out about 2640 kg of carbon dioxide in a year. How many such trees are needed to soak up all the carbon dioxide from one car in a year? [\[Forests\]](#)

- (A) 12 trees
- (B) 60 trees
- (C) 120 trees
- (D) 100 trees
- (E) 110 trees
- (F) 1200 trees
- (G) 132 trees

26. A district has a population of 28,46,150. For a quick report this is rounded to 28 lakh. By how much does the actual population differ from this rounded figure? [\[Large Numbers\]](#)

- (A) 45,150
- (B) 4,61,500
- (C) 46,850
- (D) 36,150
- (E) 46,050
- (F) 46,150

27. An Arctic tern flies about 40,000 km on its round-trip migration each year. Over a lifetime of 25 years, about how many kilometres does it fly in total? [\[Weather, Climate & Adaptations\]](#)

- (A) 10,00,000 km
- (B) 65,000 km
- (C) 1,00,00,000 km
- (D) 4,00,000 km
- (E) 16,00,000 km
- (F) 6,40,000 km
- (G) 1,00,000 km

28. What is 50 minus 23.756? [\[A Peek Beyond the Point\]](#)

- (A) 33.756
- (B) 26.254
- (C) 26.344
- (D) 26.144
- (E) 26.244
- (F) 27.244
- (G) 26.246

29. A forest covers 12500 hectares. Each year clearing destroys 4% of whatever forest is left at the start of that year. How many hectares of forest are cleared during the first two years altogether? [\[Forests\]](#)

- (A) 480 hectares
- (B) 1000 hectares
- (C) 11500 hectares
- (D) 980 hectares
- (E) 800 hectares
- (F) 11520 hectares
- (G) 500 hectares

30. Estimate 612×489 by first rounding each number to the nearest hundred. By how much does this estimate differ from the exact product? [\[Large Numbers\]](#)

- (A) 832
- (B) 268
- (C) 268000
- (D) 700
- (E) 1000
- (F) 732
- (G) 632

31. Trees in a tropical rainforest often have leaves with pointed 'drip tips'. How does this shape help the plant in its very rainy climate? [\[Weather, Climate & Adaptations\]](#)

- (A) The tips make the leaf heavier so it does not blow away
- (B) The tips help the leaf catch more sunlight
- (C) The tips scare away leaf-eating insects
- (D) The pointed tips let heavy rainwater run off the leaf quickly so it does not sit and rot the leaf
- (E) The tips draw water up from the soil faster
- (F) The tips help the plant float during floods
- (G) The tips collect rainwater and store it inside the leaf

32. A walking path is 2.5 km long, and a side path adds another 350 m. What is the total length of both paths in metres? [\[A Peek Beyond the Point\]](#)

- (A) 2535 m
- (B) 3500 m
- (C) 600 m
- (D) 285 m
- (E) 2850 m
- (F) 2850.5 m
- (G) 2350 m

33. In a forest food web there are 80 animals. Three-quarters of them are plant-eaters (herbivores) and the rest are flesh-eaters (carnivores). Of the carnivores, exactly half are top predators. How many top predators are there? [\[Forests\]](#)

- (A) 15
- (B) 10
- (C) 8
- (D) 5
- (E) 40
- (F) 20
- (G) 30

34. Write the number '5 crore' in figures. How many zeros does it have? [\[Large Numbers\]](#)

- (A) 5 zeros
- (B) 6 zeros
- (C) 7 zeros
- (D) 8 zeros
- (E) 10 zeros
- (F) 9 zeros
- (G) 4 zeros

35. A city receives an average of 4.2 cm of rain each month. About how much rain does it receive in a whole year? [\[Weather, Climate & Adaptations\]](#)

- (A) 504 cm
- (B) 48 cm
- (C) 50.4 cm
- (D) 52.4 cm
- (E) 4.2 cm
- (F) 16.2 cm
- (G) 42 cm

36. We know that $\frac{1}{8} = 0.125$. Using this idea, what is $\frac{5}{8}$ written as a decimal? [\[A Peek Beyond the Point\]](#)

- (A) 0.625
- (B) 0.58
- (C) 0.8
- (D) 0.685
- (E) 0.125
- (F) 0.575
- (G) 0.65

37. A single tree gives off about 200 litres of water vapour into the air each day. A grove of 45 such trees keeps this up every day for 30 days. How many litres of water vapour does the grove release in those 30 days? [\[Forests\]](#)

- (A) 270000 litres
- (B) 9000 litres
- (C) 2700000 litres
- (D) 45000 litres
- (E) 135000 litres
- (F) 27000 litres
- (G) 6000 litres

38. Which of these numbers is the largest: 4,50,00,000 or 45,00,000 or 4,05,00,000 or 40,50,000? [\[Large Numbers\]](#)

- (A) 40,50,000
- (B) they are all equal
- (C) 40,50,000 is largest
- (D) 45,00,000
- (E) 4,50,00,000
- (F) 4,05,00,000
- (G) 4,05,00,000 and 4,50,00,000 are equal

39. On most clear days, at roughly what time does the air temperature reach its highest, and when is it lowest? [\[Weather, Climate & Adaptations\]](#)

- (A) Highest at sunrise and lowest in the afternoon
- (B) The temperature stays the same all day and night
- (C) Highest in the early afternoon and lowest around sunrise
- (D) Highest at midnight and lowest at noon
- (E) Highest just after sunset and lowest at midday
- (F) Highest at sunset and lowest at noon

40. Round the number 7.4962 to two decimal places. [\[A Peek Beyond the Point\]](#)

- (A) 7.50
- (B) 8.00
- (C) 7.55
- (D) 7.49
- (E) 7.496
- (F) 7.40
- (G) 7.46

41. Why does the floor of a thick forest stay dark and damp even on a bright, hot afternoon? [\[Forests\]](#)

- (A) Decomposers cool the floor by releasing cold gases
- (B) The dense canopy blocks most sunlight from reaching the floor and also slows down evaporation underneath
- (C) The herb layer reflects all the sunlight back upward
- (D) Sunlight is too weak to reach the ground in any forest
- (E) Tree roots pump cold water up to keep the floor wet
- (F) Forest soil is naturally black, so it stays cool
- (G) The forest floor makes its own shade by giving off cold air

42. Add 12,34,567 and 7,65,433. What is the sum? [\[Large Numbers\]](#)

- (A) 19,99,990
- (B) 21,00,000
- (C) 20,00,000
- (D) 2,00,000
- (E) 20,10,000
- (F) 19,99,000
- (G) 20,00,100

43. A camel is well adapted to hot, dry deserts. Which statement about a camel's adaptations is correct?

[\[Weather, Climate & Adaptations\]](#)

- (A) Its hump fills with air to help it float on sand
- (B) It stores pure water in its hump for the dry season
- (C) It has webbed feet to swim across desert rivers
- (D) It hibernates underground through the hot summer
- (E) It stores fat in its hump and can go a long time without water, helping it survive where food and water are scarce
- (F) It sweats heavily all day to keep its body cool
- (G) Its thick fur is mainly to keep it warm at the equator

44. You buy 1.5 kg of apples at Rs 80 per kg and 0.75 kg of grapes at Rs 120 per kg. What is the total cost? [\[A Peek Beyond the Point\]](#)

- (A) Rs 195
- (B) Rs 240
- (C) Rs 210
- (D) Rs 205
- (E) Rs 220
- (F) Rs 21
- (G) Rs 200

45. A fallen log rots away completely in 5 years, while a fallen leaf rots completely in just 2 months. How many times faster, in terms of time taken, does the leaf rot compared with the log? [\[Forests\]](#)
- (A) 2.5 times
 - (B) 60 times
 - (C) 5 times
 - (D) 15 times
 - (E) 25 times
 - (F) 30 times
 - (G) 12 times
46. A publisher has 9,60,000 books to pack, with exactly 24 books in each carton. How many cartons are needed? [\[Large Numbers\]](#)
- (A) 38,400
 - (B) 4,400
 - (C) 4,000
 - (D) 4,00,00,000
 - (E) 44,000
 - (F) 40,000
 - (G) 4,00,000
47. On three hot days a town records temperature ranges (max minus min) of 12 deg C, 9 deg C and 15 deg C. What is the average daily temperature range over these three days? [\[Weather, Climate & Adaptations\]](#)
- (A) 10 deg C
 - (B) 9 deg C
 - (C) 18 deg C
 - (D) 11 deg C
 - (E) 12 deg C
 - (F) 36 deg C
48. What is 3.14 multiplied by 1000? [\[A Peek Beyond the Point\]](#)
- (A) 31.4
 - (B) 3140
 - (C) 314
 - (D) 0.00314
 - (E) 3140.0 hundredths
 - (F) 3.140
49. In a forest, 5000 plants support 500 plant-eating insects, which support 50 frogs, which support 5 snakes. What is the ratio of the number of plants to the number of snakes, in simplest form? [\[Forests\]](#)
- (A) 5000 : 50
 - (B) 1 : 1000
 - (C) 10 : 1
 - (D) 50 : 1
 - (E) 100 : 1
 - (F) 1000 : 1

50. Which of these is the correct expanded form of the number 6,07,089? [\[Large Numbers\]](#)

- (A) $6,00,000 + 7,000 + 8 + 9$
- (B) $6,000 + 700 + 80 + 9$
- (C) $6,00,000 + 7,000 + 80 + 9$
- (D) $6,00,000 + 7,000 + 800 + 90$
- (E) $6,00,000 + 700 + 80 + 9$
- (F) $60,000 + 7,000 + 80 + 9$
- (G) $6,00,000 + 70,000 + 800 + 9$

51. Air at a certain temperature can hold at most 30 grams of water vapour per cubic metre. Right now it actually holds 21 grams per cubic metre. What is the relative humidity of the air? [\[Weather, Climate & Adaptations\]](#)

- (A) 51%
- (B) 30%
- (C) 0.7%
- (D) 70%
- (E) 143%
- (F) 21%
- (G) 9%

52. What is the mean (average) of the four decimals 2.4, 3.6, 1.8 and 4.2? [\[A Peek Beyond the Point\]](#)

- (A) 12.0
- (B) 3.0
- (C) 3.2
- (D) 2.4
- (E) 3.5
- (F) 2.8

53. On a bare hillside the soil washed away is 12.5 tonnes per hectare each year. On the same slope when it is forested, only 0.8 tonnes per hectare is washed away each year. How many times more soil is lost from the bare slope than from the forested slope? [\[Forests\]](#)

- (A) 1562.5 times
- (B) 11.7 times
- (C) 12.5 times
- (D) 15.625 times
- (E) 16 times
- (F) 13.3 times
- (G) 20 times

54. Work out the value of (1000×1000) minus (999×999) . [\[Large Numbers\]](#)

- (A) 2001
- (B) 1
- (C) 1999
- (D) 999
- (E) 2000
- (F) 1000
- (G) 1998

55. Which of these statements about polar animals is TRUE? [\[Weather, Climate & Adaptations\]](#)

- (A) Polar bears live only in Antarctica with the penguins
- (B) Penguins live in the cold far south (Antarctic region), while polar bears live in the cold far north (Arctic)
- (C) Both penguins and polar bears live mainly in rainforests
- (D) Polar bears migrate to the equator every winter
- (E) Penguins are found mainly in hot tropical deserts
- (F) Penguins live in the Arctic and polar bears in the Antarctic

56. Which is greater, 0.7 or $\frac{3}{4}$, and by how much when written as a decimal? [\[A Peek Beyond the Point\]](#)

- (A) 0.7 is greater, by 0.05
- (B) they are equal
- (C) $\frac{3}{4}$ is greater, by 0.05
- (D) 0.7 is greater, by 0.04
- (E) $\frac{3}{4}$ is greater, by 0.25
- (F) 0.7 is greater, by 0.3

57. During daylight a tree makes 30 units of oxygen each hour for 12 hours of sunlight. But the tree breathes all 24 hours, using up 8 units of oxygen each hour. What is the tree's net oxygen given to the air in one full day? [\[Forests\]](#)

- (A) 192 units
- (B) 216 units
- (C) 264 units
- (D) 360 units
- (E) 96 units
- (F) 168 units

58. A government scheme shares Rs 2,40,00,000 equally among 1600 villages. How much does each village receive? [\[Large Numbers\]](#)

- (A) Rs 24,000
- (B) Rs 1,50,000
- (C) Rs 15,000
- (D) Rs 16,000
- (E) Rs 1,500.00
- (F) Rs 1,500
- (G) Rs 12,000

59. A region is hot throughout the year and receives very heavy rainfall in almost every month, supporting dense, tall forests. Which type of climate and region does this best describe? [\[Weather, Climate & Adaptations\]](#)

- (A) A hot, dry desert climate like the Sahara
- (B) A cold mountain climate with heavy snowfall
- (C) A cold polar climate like Antarctica
- (D) A dry grassland with one short rainy season
- (E) A tropical rainforest climate, such as the Amazon or the Western Ghats
- (F) A temperate climate with four distinct seasons

60. A jug holds 2.5 litres of juice. You pour out 0.85 litres twice. How much juice is left in the jug? [\[A Peek Beyond the Point\]](#)

[Beyond the Point\]](#)

- (A) 0.65 litres
- (B) 1.7 litres
- (C) 1.65 litres
- (D) 0.8 litres
- (E) 0.9 litres
- (F) 1.8 litres
- (G) 0.75 litres

61. In a forest, what is the main job of microorganisms such as bacteria and fungi living in the soil? [\[Forests\]](#)

- (A) They mainly hunt and eat small forest animals
- (B) They make their own food using sunlight like green leaves
- (C) They carry seeds from one part of the forest to another
- (D) They give out oxygen for animals to breathe
- (E) They break down dead plants and animals into simple nutrients that living plants can use again
- (F) They protect trees by killing all insects nearby
- (G) They store rainwater deep underground for dry seasons

62. About how many days is one million (10,00,000) seconds, rounded to one decimal place? (Take 1 day = 86,400 seconds.) [\[Large Numbers\]](#)

- (A) About 27.8 days
- (B) About 116 days
- (C) About 16.7 days
- (D) About 11.6 days
- (E) About 11.6 hours
- (F) About 1.16 days
- (G) About 6.9 days

63. In the mountains the air temperature falls by about 6.5 deg C for every 1000 metres you climb. If it is 30 deg C at sea level, about what is the temperature at a height of 2000 metres? [\[Weather, Climate & Adaptations\]](#)

- (A) 19.5 deg C
- (B) 4 deg C
- (C) 23.5 deg C
- (D) 17.5 deg C
- (E) 13 deg C
- (F) 17 deg C

64. What is the value of 1.25×0.8 ? [\[A Peek Beyond the Point\]](#)

- (A) 10.0
- (B) 0.10
- (C) 0.8
- (D) 100
- (E) 1.0
- (F) 1.25
- (G) 0.01

65. A large forest tree has a round, leafy crown that is roughly a circle of radius 7 metres. Taking pi as $\frac{22}{7}$, what is the area of ground shaded by this crown? [\[Forests\]](#)

- (A) 154 square metres
- (B) 49 square metres
- (C) 144 square metres
- (D) 308 square metres
- (E) 44 square metres
- (F) 22 square metres
- (G) 77 square metres

66. Write the number 'four crore six lakh nine hundred' in figures. [\[Large Numbers\]](#)

- (A) 4,00,60,900
- (B) 4,06,09,000
- (C) 40,06,00,900
- (D) 4,06,00,900
- (E) 4,06,000,900
- (F) 40,60,900
- (G) 4,60,00,900

67. In much of India, strong moist winds blow in from the sea between June and September and bring most of the year's rain. What is this seasonal wind system called? [\[Weather, Climate & Adaptations\]](#)

- (A) A desert dust storm
- (B) The autumn dry spell
- (C) The winter snowstorm
- (D) A daily land breeze
- (E) A coastal sea breeze that lasts one hour
- (F) The summer (south-west) monsoon
- (G) A polar cold wave

68. A ribbon 6.3 metres long is cut into pieces each 0.45 metres long. How many whole pieces can be cut?

[\[A Peek Beyond the Point\]](#)

- (A) 16
- (B) 14.5
- (C) 14
- (D) 12
- (E) 15
- (F) 140
- (G) 13

69. A forest fire spreads steadily at 3.5 metres every minute. A strip of forest is 1.4 kilometres wide. How many minutes will the fire take to burn right across this strip? [\[Forests\]](#)

- (A) 40 minutes
- (B) 400 minutes
- (C) 490 minutes
- (D) 350 minutes
- (E) 140 minutes
- (F) 200 minutes
- (G) 4000 minutes

70. A printed book has 4,80,000 words spread evenly over 600 pages. At that rate, how many words are on 45 pages? [\[Large Numbers\]](#)

- (A) 27,000
- (B) 3,600
- (C) 3,60,000
- (D) 8,00,000
- (E) 800
- (F) 36,000
- (G) 45,000

71. Over a 30-day month, the daily maximum temperatures add up to 924 deg C in total. What is the average daily maximum temperature for the month? [\[Weather, Climate & Adaptations\]](#)

- (A) 31 deg C
- (B) 30.5 deg C
- (C) 924 deg C
- (D) 30.8 deg C
- (E) 32.8 deg C
- (F) 29.8 deg C
- (G) 30 deg C

72. Write the number that is made of 2 ones, 4 tenths and 9 hundredths as a decimal. [\[A Peek Beyond the Point\]](#)

- (A) 24.9
- (B) 2.49
- (C) 9.42
- (D) 0.249
- (E) 2.094
- (F) 2.94
- (G) 4.29

73. A rainforest survey estimates about 12,50,000 living things of all kinds in one area. If four-fifths of them are insects, how many insects are there? [\[Forests\]](#)

- (A) 11,25,000
- (B) 1,00,000
- (C) 8,00,000
- (D) 12,50,000
- (E) 5,00,000
- (F) 2,50,000
- (G) 10,00,000

74. Using the digits 9, 9, 7, 2 and 0, form the greatest and the smallest 5-digit numbers possible (no number may start with 0). What is their difference? [\[Large Numbers\]](#)

- (A) 77,921
- (B) 78,921
- (C) 79,021
- (D) 79,821
- (E) 78,021
- (F) 78,911
- (G) 78,931

- 75.** A seal living in icy seas has a thick layer of fat (blubber) under its skin. What is the main purpose of this blubber? [\[Weather, Climate & Adaptations\]](#)
- (A) It insulates the body and stops heat escaping into the freezing water
 - (B) It allows the seal to glow in dark water
 - (C) It helps the seal sink quickly to the seabed
 - (D) It helps the seal digest fish faster
 - (E) It cools the seal down in the warm sun
 - (F) It is used only to make the seal look bigger
 - (G) It lets the seal breathe underwater for hours
- 76.** You pay with a Rs 500 note for items costing Rs 128.75, Rs 64.50 and Rs 199.99. How much change should you get back? [\[A Peek Beyond the Point\]](#)
- (A) Rs 106.74
 - (B) Rs 106.76
 - (C) Rs 106.86
 - (D) Rs 393.24
 - (E) Rs 107.76
 - (F) Rs 106.66
 - (G) Rs 105.76
- 77.** Suppose every decomposer in a forest suddenly disappeared. Which of these would most likely happen over the next few years? [\[Forests\]](#)
- (A) Rainfall in the forest would suddenly increase
 - (B) The soil would quickly become richer than ever before
 - (C) Dead leaves and animals would pile up and the soil would slowly lose its fertility
 - (D) Plants would grow faster because of all the dead matter
 - (E) The forest floor would become much brighter and drier
 - (F) Nothing would change, since decomposers do very little
 - (G) All the trees would immediately give off more oxygen
- 78.** A stadium has 250 rows with 184 seats in each row. If you estimate the total by rounding to 250×180 , by how much is your estimate short of the exact number of seats? [\[Large Numbers\]](#)
- (A) 46,000
 - (B) 250
 - (C) 4,000
 - (D) 184
 - (E) 45,000
 - (F) 1000
- 79.** An Arctic fox has a brownish coat in summer but grows a white coat in winter. What is the main advantage of this seasonal change of colour? [\[Weather, Climate & Adaptations\]](#)
- (A) It stays camouflaged all year - brown to match summer ground, white to match winter snow
 - (B) White fur grows because there is no food in winter
 - (C) The brown coat helps it swim in summer rivers
 - (D) The white coat keeps it cooler in winter
 - (E) The colour change scares away all predators
 - (F) The change helps the fox fly short distances

80. What is the value of 0.001 multiplied by 4,50,000? [\[A Peek Beyond the Point\]](#)

- (A) 0.45
- (B) 4.50
- (C) 4.5
- (D) 450
- (E) 4500
- (F) 45000
- (G) 45

81. A national park covers 350 square kilometres. Of this, 30% is dense forest. Of that dense forest, one-fifth is set aside as a strictly protected core zone. What is the area of the core zone? [\[Forests\]](#)

- (A) 105 square kilometres
- (B) 63 square kilometres
- (C) 21 square kilometres
- (D) 10.5 square kilometres
- (E) 42 square kilometres
- (F) 35 square kilometres
- (G) 70 square kilometres

82. In the number 88,888, how many times greater is the place value of the left-most 8 than the place value of the right-most 8? [\[Large Numbers\]](#)

- (A) 100 times
- (B) 1,000 times
- (C) 8 times
- (D) 10,000 times
- (E) 9 times
- (F) 1,00,000 times

83. Cherrapunji, one of the wettest places on Earth, receives about 1.5 times as much rain as a coastal city that gets 240 cm a year. About how much rain does Cherrapunji receive in a year? [\[Weather, Climate & Adaptations\]](#)

- (A) 480 cm
- (B) 384 cm
- (C) 120 cm
- (D) 300 cm
- (E) 360 cm
- (F) 160 cm
- (G) 240 cm

84. By how much is 5.6 greater than 5.06? [\[A Peek Beyond the Point\]](#)

- (A) 5.4
- (B) 0.6
- (C) 0.46
- (D) 0.44
- (E) 1.54
- (F) 0.054
- (G) 0.54

85. When 80 mm of rain falls on a forest, the canopy catches 0.4 of it before it reaches the ground. Of the rain that does reach the ground, 0.25 runs off the surface instead of soaking in. How many millimetres of rain run off the surface? [\[Forests\]](#)

- (A) 8 mm
- (B) 12 mm
- (C) 19.2 mm
- (D) 20 mm
- (E) 24 mm
- (F) 48 mm

86. Subtract 7,86,549 from 10,00,000. What is the result? [\[Large Numbers\]](#)

- (A) 2,13,441
- (B) 2,13,451
- (C) 3,13,451
- (D) 2,14,451
- (E) 2,13,551
- (F) 2,13,461

87. At a polar research station the temperature is -28 deg C at night and -13 deg C in the afternoon. By how many degrees does the temperature rise from night to afternoon? [\[Weather, Climate & Adaptations\]](#)

- (A) 21 deg C
- (B) 41 deg C drop
- (C) 41 deg C
- (D) -15 deg C
- (E) 15 deg C
- (F) 11 deg C

88. A rectangular garden bed measures 3.5 metres by 2.4 metres. What is its area? [\[A Peek Beyond the Point\]](#)

- (A) 8.04 square metres
- (B) 84 square metres
- (C) 5.9 square metres
- (D) 8.4 square metres
- (E) 8.6 square metres
- (F) 11.8 square metres
- (G) 6.0 square metres

89. Which of the following is NOT a way that forests help us and the environment? [\[Forests\]](#)

- (A) They hold soil together with their roots
- (B) They give us wood, fruits, medicines and other products
- (C) They speed up soil erosion and make floods worse
- (D) They slow down rainwater and reduce flooding
- (E) They provide homes for countless plants and animals
- (F) They help bring rainfall and cool the air

- 90.** A printing press prints 18,500 sheets every hour. It runs 7.5 hours a day for 6 days. How many sheets does it print in total? [\[Large Numbers\]](#)
- (A) 83,250
 - (B) 9,25,000
 - (C) 8,33,500
 - (D) 1,38,750
 - (E) 1,11,000
 - (F) 8,32,050
 - (G) 8,32,500
- 91.** Different instruments measure different weather elements. Which instrument is used to measure the humidity (amount of water vapour) in the air? [\[Weather, Climate & Adaptations\]](#)
- (A) A wind vane
 - (B) A hygrometer
 - (C) A thermometer
 - (D) A rain gauge
 - (E) A measuring cylinder
 - (F) A compass
- 92.** What is the value of (0.2) multiplied by itself three times, that is $0.2 \times 0.2 \times 0.2$? [\[A Peek Beyond the Point\]](#)
- (A) 0.08
 - (B) 0.6
 - (C) 0.8
 - (D) 0.0008
 - (E) 0.06
 - (F) 0.008
- 93.** A forest has 1200 deer. Because food is plentiful, the deer number grows by one-quarter each year. If this continues, how many deer will there be after 2 years? [\[Forests\]](#)
- (A) 2400
 - (B) 1800
 - (C) 1500
 - (D) 1650
 - (E) 1875
 - (F) 2100
 - (G) 1620
- 94.** Round the number 9,76,54,321 to the nearest crore. By how much does this rounded value differ from the actual number? [\[Large Numbers\]](#)
- (A) 23,45,679
 - (B) 23,45,779
 - (C) 13,45,679
 - (D) 2,34,56,790
 - (E) 76,54,321
 - (F) 23,45,689
 - (G) 23,54,679

- 95.** Many birds migrate thousands of kilometres each year. What is the main reason birds migrate between regions as the seasons change? [\[Weather, Climate & Adaptations\]](#)
- (A) Because their feathers fall off if they stay in one place
 - (B) To escape from the gravity of their home region
 - (C) To make their long-distance flying muscles weaker
 - (D) Because they cannot see in their home region's sunlight
 - (E) To avoid drinking any water during the year
 - (F) To reach places where the weather is suitable and food is plentiful for feeding and breeding
 - (G) Because they are pushed along helplessly by all winds
- 96.** A car uses 5.5 litres of petrol to travel 100 km. How much petrol does it need to travel 240 km at the same rate? [\[A Peek Beyond the Point\]](#)
- (A) 11.0 litres
 - (B) 132 litres
 - (C) 12.2 litres
 - (D) 13.2 litres
 - (E) 13.0 litres
 - (F) 14.2 litres
- 97.** In a food chain grass to deer to tiger, only one-tenth of the food energy passes to the next level each step. To build up 2 kg of tiger, how many kilograms of grass must the food chain start with? [\[Forests\]](#)
- (A) 2 kg
 - (B) 100 kg
 - (C) 20 kg
 - (D) 400 kg
 - (E) 2000 kg
 - (F) 22 kg
 - (G) 200 kg
- 98.** A company sold 9,87,654 units last year and 12,34,567 units this year. How many MORE units did it sell this year, rounded to the nearest thousand? [\[Large Numbers\]](#)
- (A) 22,22,221
 - (B) 2,46,000
 - (C) 2,46,913
 - (D) 2,47,000
 - (E) 2,47,913
 - (F) 2,46,900
 - (G) 2,50,000
- 99.** A hill station records its daily highest temperatures over four days as 18.5 deg C, 17.2 deg C, 19.8 deg C and 16.5 deg C. What is the mean of these four readings, to one decimal place? [\[Weather, Climate & Adaptations\]](#)
- (A) 72.0 deg C
 - (B) 19.0 deg C
 - (C) 18.2 deg C
 - (D) 18.0 deg C
 - (E) 18.5 deg C
 - (F) 17.5 deg C
 - (G) 17.0 deg C

100. Petrol costs Rs 102.50 per litre. What is the cost of 12.5 litres? [\[A Peek Beyond the Point\]](#)

- (A) Rs 1025.00
- (B) Rs 1282.25
- (C) Rs 1312.50
- (D) Rs 1280.25
- (E) Rs 1281.50
- (F) Rs 1218.25
- (G) Rs 1281.25